FL-05 — Agent Concepts and MCP Basics

Assignment Code: FL-05
Track: General AI Fluency
Week: 4
Phase: Build
Workload: 5 hours

1. Workflow vs Agent

What is a Workflow?

A workflow is a predefined sequence of steps used to complete a task. The steps, order, inputs, and outputs are decided before the workflow runs.

What is an Agent?

An agent is an AI system that can work toward a goal by deciding which actions or tools to use and what steps to take. Instead of following only one fixed sequence, it can choose actions based on the situation and the information it receives.

My FL-04 Workflow

My FL-04 pipeline is a **workflow**.

The pipeline follows a predefined process:

Input → Draft → Critique → Revise → Human Review

The sequence was designed before running the workflow. Each step has a defined purpose and passes its output to the next step. The AI does not independently decide which steps to perform.

Workflow vs Agent Comparison

Workflow	Agent
Follows predefined steps	Can decide which steps to take
Sequence is designed beforehand	Sequence can change based on the situation
Has defined handoffs	Can select actions/tools as needed
More predictable	More flexible
Easier to control	Requires more monitoring

2. What is MCP?

MCP stands for Model Context Protocol.

MCP provides a standardized way for AI applications to connect with external tools and sources of information.

Instead of an AI working only with information directly provided in a conversation, MCP can allow it to interact with external systems through connected servers.

MCP Primitives

2.1 Tools

Tools allow an AI to perform actions through an external system.

Examples include:

- Reading a file
- Searching information
- Querying a service
- Creating or modifying information
- Performing an operation in another application

In simple: A tool is something the AI can use to perform an action.

2.2 Resources

Resources provide information that an AI can access through an MCP connection.

Examples include:

- Files
- Documents
- Database information
- Other external information sources

In simple: A resource is information the AI can access.

2.3 Prompts

MCP prompts provide reusable instructions or templates that help guide interactions with an MCP-connected system.

In simple: A prompt provides structured guidance for using the connected system.

Summary

Tools = actions
Resources = information
Prompts = reusable instructions

3. MCP / Connector Setup

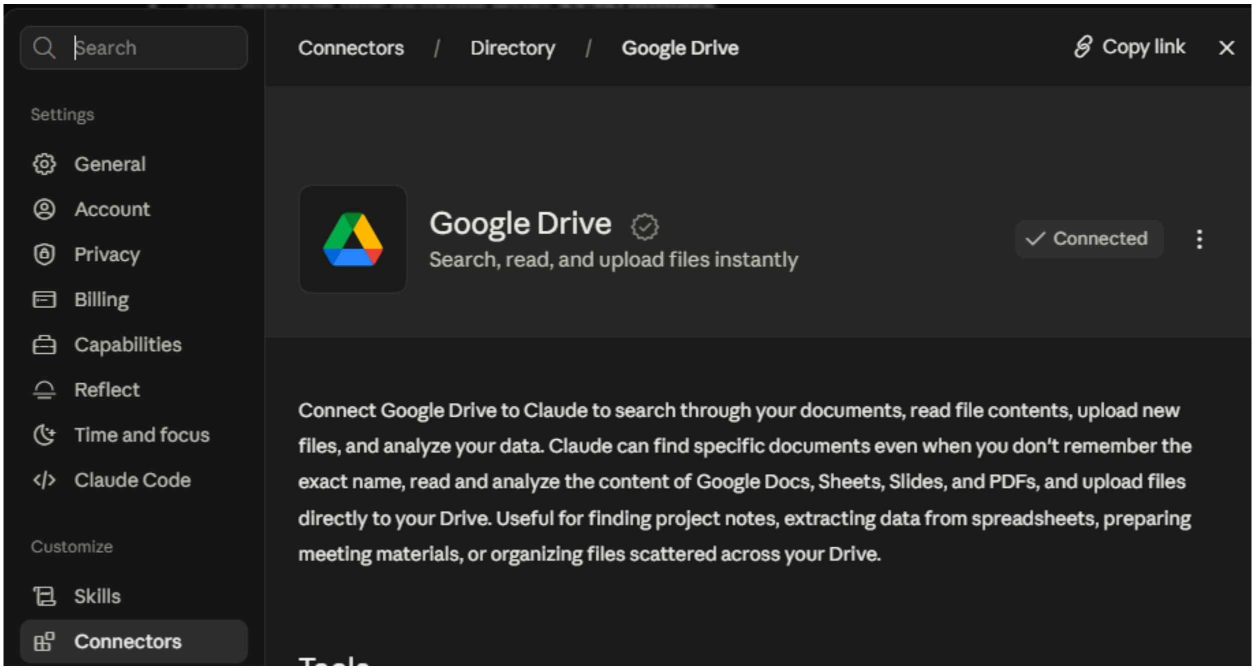
MCP Server or Connector Used:
Google Drive MCP Server

AI Client:
Claude

Purpose of the Connection:
The Google Drive connector allows Claude to access your cloud-based files, retrieve specific document text, inspect folder structures, and reference up-to-date working notes or spreadsheets directly during a session. This transforms Claude from a static conversational assistant into an active retrieval client that can pull real-time context from your personal or shared workspace.

Setup:
I connected the Google Drive MCP Server to Claude so that Claude could access files stored in my Google Drive through the connected tools. I used the connection to retrieve and work with information from my Drive instead of relying only on information provided directly in the chat.

Evidence of Setup



4. MCP Task 1

Task

The task is to Read the project file and identify the main sections.

Why Chat Alone Could Not Do This

Chat alone don’t have access to my Google Drive files that is why I have to use Google Drive MCP Connector

Tool Used

Google Drive MCP Server

Result

I found two files in your Drive with this exact name — I read the one you most recently viewed (dated Nov 2022, viewed just now). Here's what's in it:

File: "Copy of DA OnRamp - Starter Project" (Google Sheet)

This is a single-sheet spreadsheet, not a multi-section document — it contains one data table titled "User Data" with no other tabs, headers, or narrative sections. Its structure:

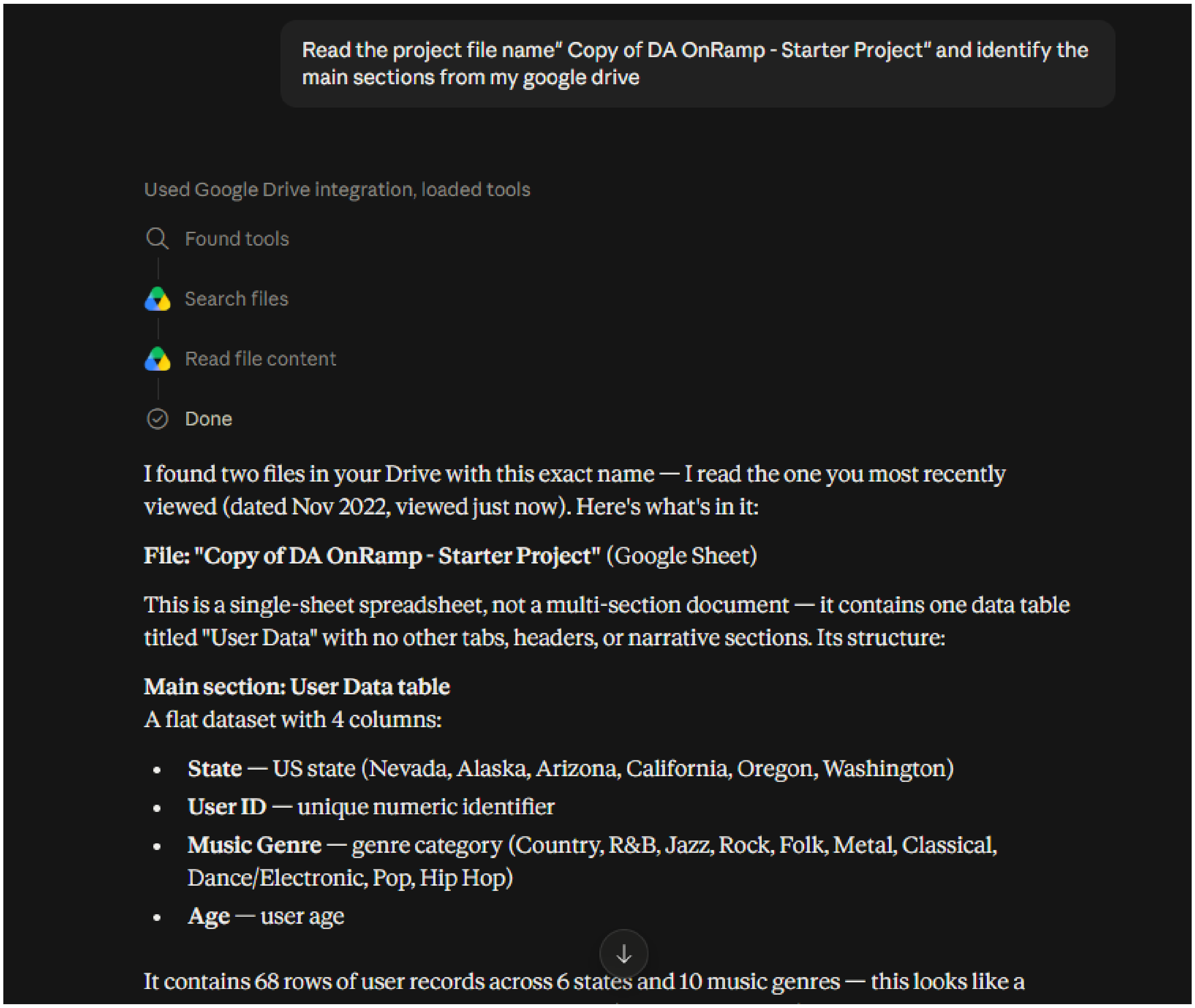
Main section: User Data table
A flat dataset with 4 columns:

- **State** — US state (Nevada, Alaska, Arizona, California, Oregon, Washington)
- **User ID** — unique numeric identifier
- **Music Genre** — genre category (Country, R&B, Jazz, Rock, Folk, Metal, Classical, Dance/Electronic, Pop, Hip Hop)
- **Age** — user age

It contains 68 rows of user records across 6 states and 10 music genres — this looks like a sample dataset for a data analytics onboarding/practice exercise (the "DA OnRamp" name suggests a Data Analytics onboarding starter file), likely meant for practicing filtering, grouping, or pivot-table type analysis rather than being a document with multiple written sections.

Note: there's a second file with the identical name (created earlier the same day, Nov 2022) containing the same data sorted differently. Let me know if you'd like me to check that one too, or if you want help analyzing this data (e.g., breakdowns by state or genre).

Evidence



5. MCP Task 2

Task

Find all references to the state "Nevada" in the connected files

Why Chat Alone Could Not Do This

Chat alone don’t have access to my Google Drive files that is why I have to use Google Drive MCP Connector

Tool Used

Google Drive MCP Server

Result

Here are all the Nevada references in the "Copy of DA OnRamp - Starter Project" file:

State	User ID	Music Genre	Age
Nevada	12345678	Country	40
Nevada	10123455	Classical	30
Nevada	11234566	Dance/Electronic	42
Nevada	21234565	Classical	31
Nevada	31234564	Metal	45
Nevada	43456785	Classical	33
Nevada	46790118	R&B	53
Nevada	47901229	Folk	29
Nevada	55679006	R&B	50
Nevada	56790117	Pop	34
Nevada	63456783	Country	45
Nevada	77901226	Dance/Electronic	40

Total: 12 Nevada records, spanning 7 genres:

- Classical (3): ages 30, 31, 33
- Country (2): ages 40, 45
- Dance/Electronic (2): ages 40, 42
- R&B (2): ages 50, 53
- Folk (1): age 29
- Metal (1): age 45
- Pop (1): age 34

Age range: 29–53 (average ≈ 39.4)

Note: this is from the file you most recently viewed. There's a second identically-named file (same data, different sort order) — let me know if you'd like me to check that one too, though the Nevada records should be identical since it's the same dataset.

Evidence

Find all references to the state "Nevada" in the connected files

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- Classical (3): ages 30, 31, 33
- Country (2): ages 40, 45
- Dance/Electronic (2): ages 40, 42
- R&B (2): ages 50, 53

6. MCP Task 3

Task

Compare the information in these two connected files and summarize the differences.

Why Chat Alone Could Not Do This

Chat alone don't have access to my Google Drive files that is why I have to use Google Drive MCP Connector

Tool Used

Google Drive MCP Server

Result

The two files contain **identical data** — no differences in content.

Comparison summary:

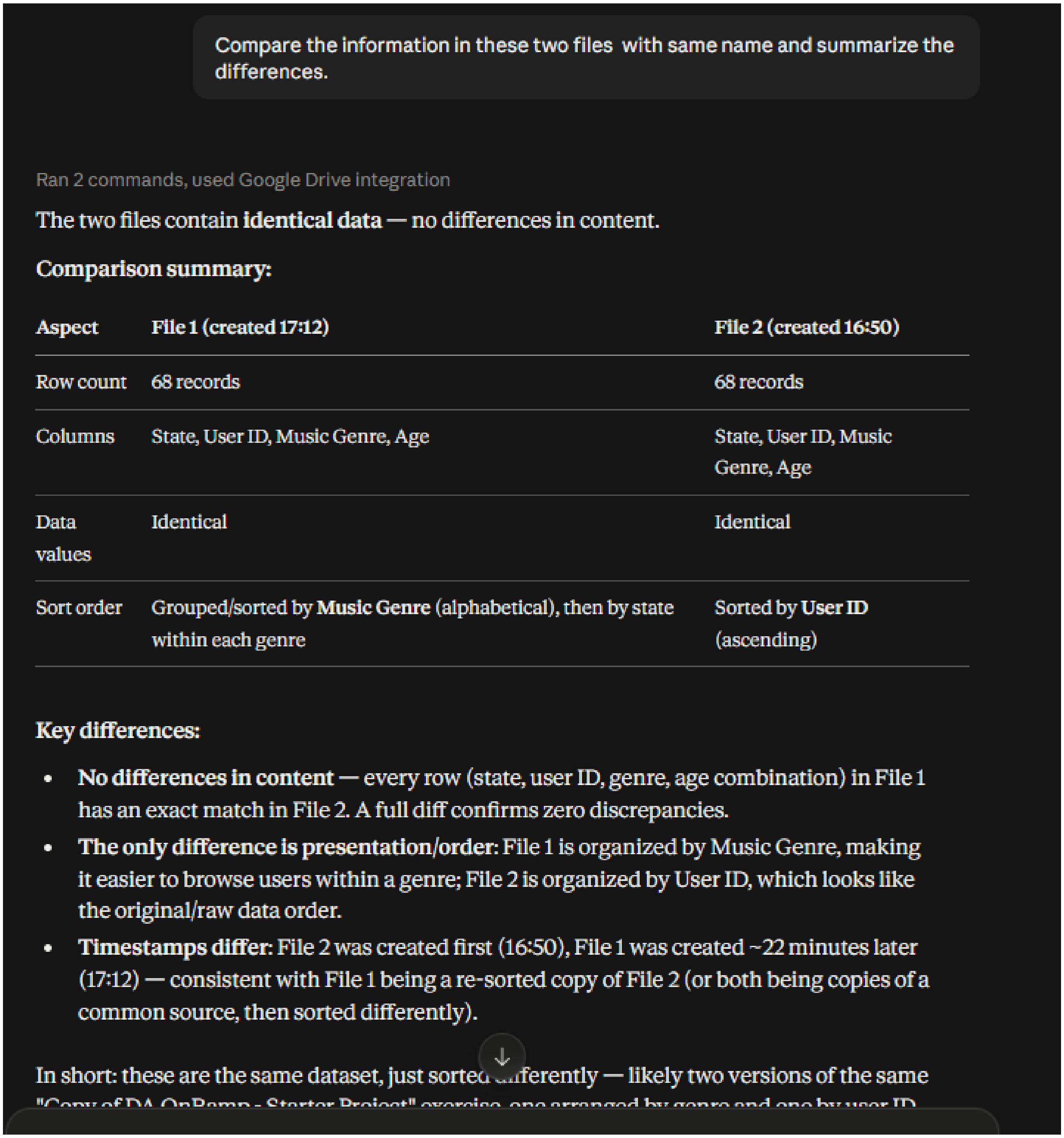
Aspect	File 1 (created 17:12)	File 2 (created 16:50)
Row count	68 records	68 records
Columns	State, User ID, Music Genre, Age	State, User ID, Music Genre, Age
Data values	Identical	Identical
Sort order	Grouped/sorted by **Music Genre** (alphabetical), then by state within each genre	Sorted by **User ID** (ascending)

Key differences:

- **No differences in content** — every row (state, user ID, genre, age combination) in File 1 has an exact match in File 2. A full diff confirms zero discrepancies.
- **The only difference is presentation/order**: File 1 is organized by Music Genre, making it easier to browse users within a genre; File 2 is organized by User ID, which looks like the original/raw data order.
- **Timestamps differ**: File 2 was created first (16:50), File 1 was created ~22 minutes later (17:12) — consistent with File 1 being a re-sorted copy of File 2 (or both being copies of a common source, then sorted differently).

In short: these are the same dataset, just sorted differently — likely two versions of the same "Copy of DA OnRamp - Starter Project" exercise, one arranged by genre and one by user ID, with no actual data discrepancies between them.

Evidence



Compare the information in these two files with same name and summarize the differences.

Ran 2 commands, used Google Drive integration

The two files contain **identical data** — no differences in content.

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7. Agent Concepts and MCP Explainer

What is an Agent?

An agent functions like a digital worker. It breaks down a large objective into smaller tasks. For example, if you ask an agent to research a topic, it does not just guess the answer. It searches the web, reads the pages, checks if it found enough facts, runs a follow-up search if needed, and then writes a final summary on its own.

What is MCP?

The Model Context Protocol (MCP) is an open standard that lets AI models securely connect to local or remote data sources and tools. Think of it as a universal USB port for artificial intelligence. It gives AI a safe and structured way to talk to databases, file systems, and software APIs without custom code for every single app.

MCP Tools, Resources and Prompts

Tools:
Executable functions that the AI can run to change state or take an action, such as sending an email, running code, or querying a database.

Resources:
Data sources that the AI can read to gather context, such as text files, API payloads, or document repositories.

Prompts:
Pre-defined templates or starter instructions provided by the server to guide how the AI interacts with specific data or workflows.

Workflow vs Agent

My FL-04 pipeline is a workflow because:

It moves in a strict, straight line from Input to Draft, Critique, Revise, and Human Review. Every step runs every single time in the exact same order.

An agent version would be different because:

It would evaluate the text during the process and dynamically skip or repeat steps based on quality. If the draft is already great, it might skip extra revisions. If the critique finds major gaps, it might trigger extra research instead of moving straight to human review.

What Would My FL-04 Workflow Need to Become an Agent?

It would need a decision-making loop powered by an LLM that controls the flow. Instead of a hard-coded pipeline, the system would inspect the draft and ask: "Is this ready?" If the score is low, it calls the research tool using MCP to get more facts, rewrites the draft, and tests it again. It only routes the task to human review when self-evaluation confirms the criteria are met.

8. One Concrete Agent Upgrade

Current FL-04 workflow:

Input → Draft → Critique → Revise → Human Review

Proposed agent upgrade:

Replace the static Revise step with a conditional loop where the AI evaluates its own draft against a rubric and decides whether to write a new version, fetch missing data via MCP, or pass the file forward.

Example

The system could receive the goal of creating a final piece of content and decide whether it needs additional information, inspect connected resources when necessary, create a draft, evaluate the draft, revise it when required, and stop when the result meets the defined requirements.

Why This Makes It More Agent-Like

It shifts control from a rigid script to the AI itself. The system actively decides what action to take next based on intermediate results rather than blindly running every predefined node in a fixed line.

9. Evidence Checklist

- ☑ MCP server or connector successfully connected
- ☑ Setup screenshot included
- ☑ Three different MCP/connector tasks completed
- ☑ Task 1 screenshot included
- ☑ Task 2 screenshot included
- ☑ Task 3 screenshot included
- ☑ Screenshots show actual tool calls/results
- ☑ Workflow vs agent distinction explained

- ☑ FL-04 correctly classified as a workflow
- ☑ MCP explained in my own words
- ☑ Tools explained
- ☑ Resources explained
- ☑ Prompts explained
- ☑ One concrete agent upgrade identified
- ☑ Explainer is between 600 and 900 words